

Md. Rubiath Islam

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EDUCATION

University of Texas at Arlington, Arlington, Texas

PhD in Quantitative Biology 2026-Present

Shahjalal University of Science and Technology, Sylhet, Bangladesh

MS., with distinction in Biochemistry and Molecular Biology (3.86/4.0) 2026

BSc., with distinction in Biochemistry and Molecular Biology (3.91/4.0) 2024

RESEARCH INTEREST

Microbiome | Immune modulation | Stem cell biology | Antibiotic resistance | Pathogen-host interaction

RESEARCH EXPERIENCE

Graduate Research Assistant, Institute of Food Science and Technology, Bangladesh Council for Scientific and Industrial Research, Dhaka, Bangladesh 2025

Dissertation (Thesis): Metagenomics and integron based analysis of abundance and temporal persistence of antibiotic resistance genes in hospital wastewater.

- Investigating the abundance of species and clinically important antibiotic-resistant genes (ARGs) in hospital wastewater using high-throughput sequencing.
- Investigating the temporal persistence of the clinically important ARGs in hospital wastewater using RT-PCR.
- Evaluating the characteristics of integrons and transposons to determine the potential for persistence and dissemination of their embedded gene cassettes using Sanger sequencing.
- Learning advanced molecular biology techniques and developed metagenomics analysis pipeline.

Undergraduate Research Assistant, SUST, Sylhet, Bangladesh 2021-2024

Project-1: Co-selection pressure of heavy metals on antibiotic resistance in Enterobacteria isolated from chicken cloaca.

- Elucidated the influence of heavy metal exposure on antibiotic resistance and species-specific transcriptional reprogramming.
- Designed and optimized the study protocols, conducted RNA isolation and RT-PCR to quantify RNA.

Project-2: Multidrug and heavy metal resistant profiling of *E. coli* isolated from broiler chicken cloaca and contact surfaces from wet markets

- Investigated the prevalence of multidrug resistant and heavy metal tolerant *E. coli* from broiler chicken and wet market environment using conventional culture method and molecular techniques.
- Conducted biochemical tests, DNA isolation, and PCR to assess heavy metal and antibiotic resistance genes.

Project-3: Isolation and molecular characterization of extended-spectrum beta-lactamase-producing *E. coli* and *K. pneumoniae* from hospital setups

- Characterized β -lactamase producing *E. coli* and *K. pneumoniae* from hospital wards to assess the risk of acquiring nosocomial infection.
- Conducted biochemical identification, conventional PCR, Sanger sequencing and statistical analysis.

PUBLICATIONS

- **Islam, M.R.,** Ahmed, I., Sarkar, R., Dey, S.S., Patwary, M.R.H., Hasan, Z., Chowdhury, A., Draft genome sequence and resistome analysis of multidrug-resistant *Klebsiella pneumoniae* ST147 strain FM2_IFST_BCSIR with high mobility potential from Bangladeshi hospital wastewater in Dhaka, Bangladesh. *ASM Microbiology Resource Announcements* (Submitted, submission id: MRA00306-26) (2026)
- **Islam, M.R.,** Sarkar, R., Dey, S.S., Ahmed, I., Patwary, M.R.H., Hasan, Z., Chowdhury, A., Metagenomic profiling and temporal dynamics of antimicrobial resistance genes across hospital wastewaters in Dhaka, Bangladesh. *ASM Microbiology Spectrum* (Submitted, submission id: Spectrum00755-26) (2026)
- **Islam, M.R.,** Hossain, S., Ahmed, I., Hossain, M.M., Ritu, N.A., Mishu, M.A., Investigating the prevalence, antimicrobial resistance, and pathogenic profiles of multi drug resistant *Escherichia coli* isolated from wet markets in Bangladesh and their survival trends in heavy metals. *BMC Microbiology* (Submitted, submission id: a09a3db9-b55f-4885-82ee-f740c12b3ac8) (2025)
- **Islam, M. R.,** Maliha, A., Hossain, M. M., Ahmed, I., Hossain, M. S., Mishu, M. A., & Ghosh, A. (2026). The Role of Human Microbiome in Cancer Metastasis: Mechanisms and Therapeutic Implications. *Advances in Redox Research*, 100162. (2026)
- **Islam, M.R.,** Hossain, M.M., Mishu, M.A., Ghosh, A. Co-exposure to heavy metals and antibiotics enhances resistance and transcriptional adaptation in enteric pathogens from poultry. *Total Environment Microbiology I*, 100039 (2025).
- Mishu, M. A., Imran, A., Saha, A., Ferdousee, S., **Islam, M. R.,** Shoyeb, M.A., Islam, S., Azad, M.A.K. Detection, characterization, and antibiotic resistance profiling of multidrug-resistant bacteria isolated from circulating currency in the Northeastern region of Bangladesh. *IJID regions*, 14, 100519 (2025).
- **Islam, M. R.,** Azmal, M., Prima, F. S., Zaman B., Hossain, M.M., Ghosh, A. Retention of methicillin susceptibility in *Staphylococcus aureus* using natural adjuvant as an allosteric modifier of penicillin-binding protein 2a. *Computers in Biology and Medicine*, 181, 109070 (2024).
- **Islam, M. R.,** Rabbi, M. A., Hossain, T., Sultana, S., Uddin, S. Mechanistic approach to immunity and immunotherapy of Alzheimer's disease: A review. *ACS Chemical Neuroscience* (2024).

- Imran, M.A., **Islam, M.R.**, Saha, Ferdousee, S., Mishu, M. A., Ghosh A. Development of multi-epitope-based subunit vaccine against Crimean-Congo Hemorrhagic Fever Virus using reverse vaccinology approach. *International Journal of Peptide Research and Therapeutics* 28, 124 (2022).

PRESENTATIONS

Poster Presentation

- Islam, M.R.**, Hossain, M.M., Mishu, M.A., Ghosh, A. *Comprehensive Analysis of Different Heavy Metals Modulating Antibiotic Resistance in Enterobacteriaceae Isolated from Poultry Chicken Rectum*. International Biotechnology Conference, Brac University, 2025
- Mishu, M.A., Imran, M.A., Saha, A., **Islam, M.R.**, Ferdousee, S., Shoyeb, M.A., Azad, M.A.K. *Isolation and Molecular Characterization of β -lactam Resistant E. coli and K. pneumoniae from Hospital Settings in Sylhet*. BSBMB-SABC-OMC International Conference, University of Dhaka, Bangladesh 2023

TEACHING & MENTORING

Teaching Assistant

2024-2025

- Conducted 2 hours of lab sessions on Microbiology (BMB327L) and demonstrated basic molecular and microbiology techniques.
- Helped the instructor with class lecture preparation.
- Provided training and project oversight to a senior year student completing their Senior year project (BMB433).

FELLOWSHIPS

- National Science and Technology Fellowship**, awarded by the Ministry of Science and Technology, Government of the Republic of Bangladesh. 2025
- Competitive fellowship awarded for MS. Thesis research proposal.

AWARDS & HONORS

- Best Poster**, International Biotechnology Conference, BRAC University, Dhaka. 2025
- Best Writing**, BMB Symposium, SUST 2024
- Best Poster**, BSBMB-SABC-OMC International Conference, University of Dhaka 2023

LEADERSHIP & OUTREACH

- Education and Cultural Secretary**, Biochemistry and Molecular Biology Society, SUST. 2023-2024

- Organized quiz competitions, scientific talks, and volunteered at BMB Symposium at SUST.

Team Co-ordinator, 9th Bangladesh and 1st SAANSO Scout Jamboree

2014

- Led a team in an international scout training event to accomplish training on social welfare.

REFERENCES

Md. Zafrul Hasan, PhD. (Thesis Supervisor) – Associate Professor, Department of Biochemistry and Molecular Biology, SUST. | Email: zafrul-bmb@sust.edu

Ajit Ghosh, PhD. (Project Supervisor) – Associate Professor, Department of Biochemistry and Molecular Biology, SUST. | Email: aghosh-bmb@sust.edu